

# 光線特勤隊・任務五 透鏡 | 第 1 節 教案逐字台詞

竹東國中 自然科 | 翰林二上 4-4 透鏡與眼睛 | Ka-IV-9 生活中有許多運用光學原理的實例或儀器，例如：透鏡、眼睛、眼鏡 | po-IV-1、pe-IV-2、ai-IV-1

上課日期：**10/7**（週三）乙班第**4**節 **11:05-11:50**、甲班第**7**節 **15:00-15:45** 出席：\_\_\_\_\_

## 一、這一節在做什麼

謎題（開場丟出，結尾破解）：「放大鏡只會把東西變大嗎？」

核心概念：光進出鏡片會換方向（接任務四折射）。中間厚＝凸透鏡：看字變大、光會聚到焦點、遠處東西投影會倒過來。中間薄＝凹透鏡：看字變小、光會發散。

生活主線：放大鏡、近視眼鏡、老花眼鏡、手機鏡頭、門上貓眼。

課型：單元三節固定「體驗→操作→應用」，本節是第 1 節「體驗」：先看到、先摸到，再命名。

## 二、今日視覺流程（可直接印成流程條，完成一項翻面一項）

|                                                                                        |                                                                                        |                                                                                       |                                                                                       |                                                                                          |                                                                                          |
|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| ①                                                                                      | ②                                                                                      | ③                                                                                     | ④                                                                                     | ⑤                                                                                        | ⑥                                                                                        |
|  心情打卡 |  投影魔術 |  摸鏡片 |  看光走 |  生活闖關 |  收尾打卡 |

## 三、準備清單

| 教具                                                                                                                                                                      | 數量      | 備註                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---------------------------------|
| 放大鏡（凸透鏡）                                                                                                                                                                | 每桌 1 支  | 理化教室或文具店；越大越好                   |
| 凹透鏡 或 近視眼鏡                                                                                                                                                              | 1-2 個   | 理化教室光學組；沒有就借一副近視眼鏡              |
| A4 白紙                                                                                                                                                                   | 每桌 2 張  | 投影用、看字用                         |
| 教室窗簾                                                                                                                                                                    | 拉一半     | 窗戶要亮，教室後面要暗，投影才清楚               |
| 手電筒                                                                                                                                                                     | 1 支     | 老師示範；不照眼睛                       |
| 圓形透明水瓶 + 畫了箭頭的紙                                                                                                                                                         | 1 組（選配） | 水瓶也是凸透鏡，箭頭會反過來                  |
|   | 各印一份    | worksheet.html 列印 3 頁；第 3 頁為解答卡 |
| 完成貼紙、心情卡                                                                                                                                                                | 全班      | 固定儀式用                           |

△ 安全約定（上課前貼黑板）：放大鏡不對太陽（會燒起來）、不對眼睛；鏡片輕拿輕放。

## 四、教學流程與逐字台詞（45 分鐘）

提示層級：3 獨立 → 2 手勢／口語 → 1 部分肢體 → 0 尚未形成。提問後等待 5-10  

| 段落／時間                      | 教師逐字台詞（簡單句 → 接專有名詞）                                                                           | 學生做什麼  操作／  推理               | 帶法・提示・等待                                                                                           |
|----------------------------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| 開場儀式 3'<br>（暖身可播「光光在哪裡」MV□ | 「光線特勤隊，任務簽到！今天心情是哪一張？」（等 5 秒）<br>【正念時刻】「光之泡泡呼吸：吸——吹一個大泡泡；吐——泡泡慢慢飛走。」<br>「今天是新任務：任務五。」翻流程條第一格。 |  指心情卡、跟著吸吐。<br> 說一句「我今天——」。    | 固定儀式，不要省。<br> |
| 動機 5'<br>投影魔術              | 「大家看窗戶。」（關前面的燈）「我拿放大鏡，站在教室後面。」慢慢前後移動，讓窗戶投在白紙上。                                                |  指「正的／倒的」卡。<br> 說「我覺得——，因為——」。 | 謎題不揭曉。<br>迷思診斷：大多會說「變大」「正的」。<br>紙拿穩，讓全班輪流靠近看。                                                      |

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|                               | <p>【老師問】「紙上出現什麼？」（等 8 秒）</p> <p>「先不要說對不對，記住你的答案。今天的謎題是：放大鏡只會把東西變大嗎？」</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               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| <p>探索 10'</p> <p>摸一摸・看一看</p>  | <p>每桌發放大鏡。「用手指輕輕摸鏡片的中間，再摸旁邊。」（等 5 秒）「中間比較厚？還是比較薄？」</p> <p>再傳凹透鏡（或近視眼鏡）：「這一個摸起來呢？」</p> <p>「拿放大鏡看學習單上的字。」「換近視眼鏡看。」</p> <p>開透鏡實驗室「站 2 摸一摸」「站 4 看字」，請學生上台點。</p> <p>【鼓勵語】「你有注意到手指摸到的不一樣——這就是科學家的觀察！」</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    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| <p>建構 10'</p> <p>命名・看光走</p>   | <p>「中間厚的鏡片，叫凸透鏡。中間薄的鏡片，叫凹透鏡。」（貼名詞卡）</p> <p>「光進出鏡片會換方向——記得任務四的吸管嗎？這就是折射。」</p> <p>實驗室「站 3 光線模擬器」按開燈：「凸透鏡把光聚在一起，聚在一點，叫焦點。」「凹透鏡把光散開。」</p> <p>再開「站 1 投影魔術」按顯示光線：「樹頂的光，穿過鏡片跑到下面——所以紙上的窗戶倒過來了！」</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>● 按開燈、拉滑桿、指出焦點。</p> <p>● 說「因為光進出鏡片換方向，聚在一起，所以——」。</p>   | <p>名詞：凸透鏡、凹透鏡、會聚、發散、焦點。</p> <p>名詞卡一次一張，貼黑板固定位置。</p> <p>焦距先不深談，只說「越厚，光聚得越近」。</p> |
| <p>應用 10'</p> <p>生活闖關・學習單</p> | <p>實驗室「站 5    幫   *                                                                                                                                                                                                                                                                                                                                                                                                              </p> |                                                            |                                                                                 |

|  |                                  |  |  |
|--|----------------------------------|--|--|
|  | 完成貼紙。<br>預告：「下一次，每一組自己拿放大鏡投影窗戶！」 |  |  |
|--|----------------------------------|--|--|

## 五、工作分析：投影魔術（第 2 節學生自己做，本節老師示範）

| 步驟 | □□             | 目標提示層級      |
|----|----------------|-------------|
| 1  | 面向亮亮的窗戶，站在教室後面 | 3 獨立        |
| 2  | 一人拿白紙，一人拿放大鏡   | 2 口語        |
| 3  | 放大鏡離紙大約一個手掌寬   | 2 示範        |
| 4  | 放大鏡慢慢前後移動，眼睛看紙 | 1 部分肢體（扶手腕） |
| 5  | 紙上出現清楚的窗戶，停住   | 2 口語「停！」    |
| 6  | 說出或指出：正的還是倒的   | 3 獨立        |

## 六、IEP 目標對應（預留欄位）

| 學生代碼 | 對應 IEP 目標 | 本節觀察重點 |
|------|-----------|--------|
|      |           |        |
|      |           |        |
|      |           |        |

## 七、迷思與評量

常見迷思：放大鏡只會放大；凹透鏡沒有用；鏡片越厚就越好。

評量（實作表現）：能摸出並指出中間厚／中間薄；能分出凸透鏡與凹透鏡；能說或指出「凸透鏡讓字變大、凹透鏡讓字變小」；能說出放大鏡不能對太陽。

## 八、下兩節預告（大綱）

第 2 節 操作：投影大挑戰——預測 → 做 → 修正。每組用放大鏡把窗戶投在白紙上，找最清楚的位置、用尺量距離；換凹透鏡試試看投不投得出來；水瓶翻箭頭。

第 3 節 應用：眼睛與眼鏡——近視看遠方模糊要戴凹透鏡、老花看近處模糊要戴凸透鏡；相機、投影機；結案大搶答與安全決定。

真實連結加碼：第 6 站可開鏡頭，把真的放大鏡擋在鏡頭前拍下「變大／倒過來」當證據（沒有鏡頭就選一張照片）。

網站：任務五首頁 <https://classtools1020.github.io/light-lens/>（透鏡實驗室、學習單、代課包都在這裡）